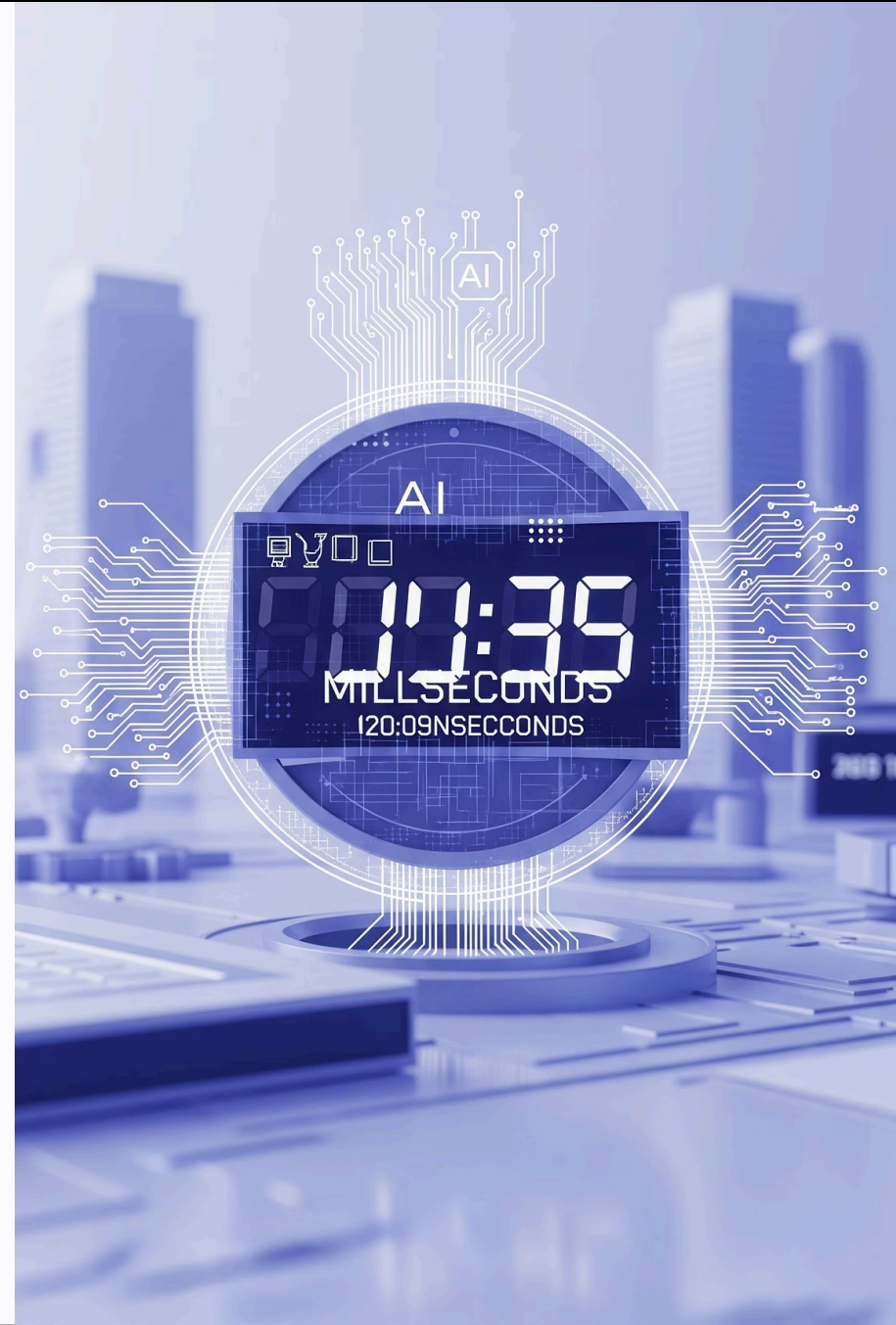


Real-Time AI Use Cases: Driving Enterprise Value Through Millisecond Decisions

An exploration of how latency-critical AI applications are transforming high-impact domains and creating billions in value



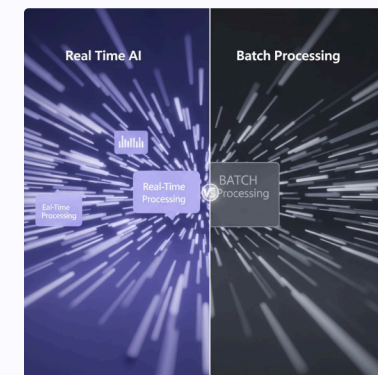
What Is Real-Time AI?

AI systems engineered to make **decisions in milliseconds**, contrasting with traditional batch AI that operates on timescales of hours or days.

These systems are powered by two critical components:

- Continuous **streaming data pipelines**
- Optimized **low-latency inference infrastructure**

Found in critical applications like ad auctions, fraud detection systems, and personalized content feeds where immediate decisions create significant value.



Why Real-Time Matters



User Expectations

Today's users expect instant, responsive experiences across all digital touchpoints



Milliseconds = Millions

Even small delays directly impact revenue in ads, banking, e-commerce and other high-value domains



Feedback Loops

Enables continuous adaptation, learning and optimization in response to changing conditions

 **Revenue Impact:** Studies show that just 100ms of added latency can reduce conversion rates by up to 7% in e-commerce and advertising platforms

Use Case: Real-Time Ads Targeting

Contextual ad auctions complete in **under 100 milliseconds**, determining:

- Which ad to display to each user
- Optimal bid amount for maximum ROI
- Placement and creative variation

These decisions incorporate user profiles, contextual signals, and historical performance data.



1

Feature Store

Millisecond retrieval of user and context data

2

Model Serving

Optimized for low-latency API responses

3

Global CDN

Fast delivery and reduced network latency

Use Case: Fraud Detection

Every financial transaction is scored in real-time with **sub-50ms latency** to prevent fraudulent activity before it completes.

Models evaluate complex patterns across:

- User history and behavior patterns
- Location and device fingerprinting
- Transaction characteristics and anomalies



Streaming Ingestion

Kafka/Flink pipelines for real-time events



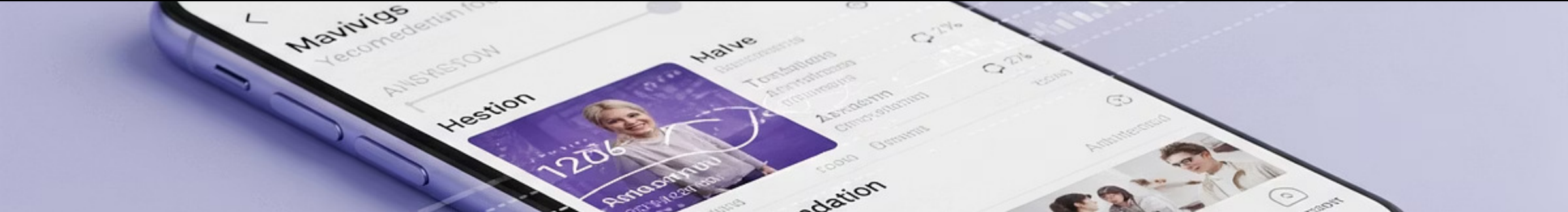
Edge Deployment

Localized scoring to minimize latency



Monitoring

Detect adversarial adaptation in real-time



Use Case: Personalization Engines

Recommendation systems that instantly update as users interact, continuously refining content and product suggestions based on behavior.



Content Feeds

TikTok and social platforms instantly adapt content based on engagement signals and viewing patterns



Streaming Services

Netflix and similar platforms dynamically update carousels and recommendations during browsing sessions



E-Commerce

Amazon and retailers adjust product recommendations in real-time based on browsing behavior and purchase intent

Infrastructure Requirements

Online Learning

Incremental model retraining

Vector Databases

Fast similarity search (Pinecone, Weaviate)

Caching Layers

Sub-millisecond access to hot features

Common Technical Requirements

10-100ms

Response Time

Maximum latency budget for real-time AI systems to deliver value

1M+

Requests/Second

Throughput needed for enterprise-scale applications

Infrastructure Requirements

- **Elastic scaling** across multiple regions to handle traffic spikes
- **Resilient architecture** with failover if model endpoints experience issues
- **Comprehensive observability** tracking model drift, latency and cost metrics

Data & Feature Infrastructure



Streaming Platforms

Kafka, Pulsar, and Kinesis provide the backbone for real-time data movement

Feature Stores

Feast and Tecton maintain consistency between online serving and offline training data

Real-time ETL

Flink, Spark Streaming, and Materialize transform raw events into ML-ready features

Event-driven APIs

Webhooks and event handlers enable immediate updates when conditions change

Model Serving Infrastructure



Low-latency Inference

Specialized servers like Triton, TensorRT, and ONNX Runtime optimize model execution for minimal latency

Experimentation

A/B testing frameworks and canary deployments enable safe production experimentation

Deployment Infrastructure

Kubernetes-based platforms like KServe and Ray Serve handle autoscaling, load balancing, and deployment automation

Edge Computing

Distributing inference to edge locations reduces network latency for time-sensitive decisions

Challenges in Real-Time AI



Latency vs. Accuracy Trade-offs

Complex models offer better accuracy but require more computation time, creating difficult engineering trade-offs



Cost Explosion

Always-on inference systems with high-throughput requirements can dramatically increase infrastructure costs



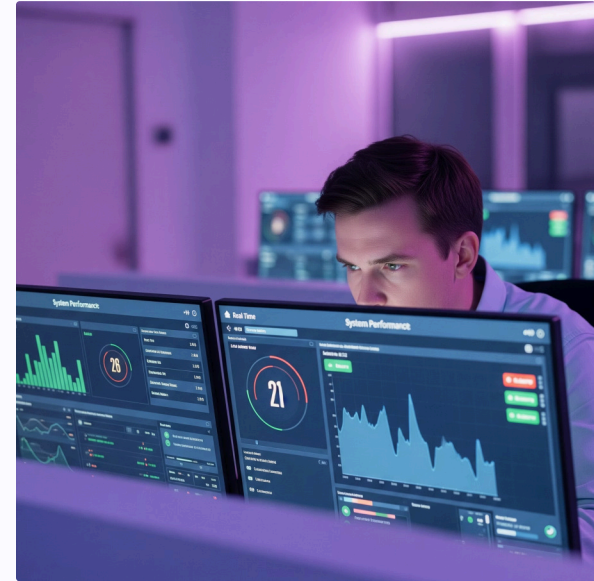
Feature Freshness

Ensuring all model inputs are updated fast enough to reflect the current state accurately



Failure Impact

Even milliseconds of delay or downtime translate directly to lost revenue or missed fraud detection



"The hardest part isn't building a model that works — it's building one that works *fast enough* and *reliably enough* for production."

Case Study: E-Commerce Personalization

The Problem

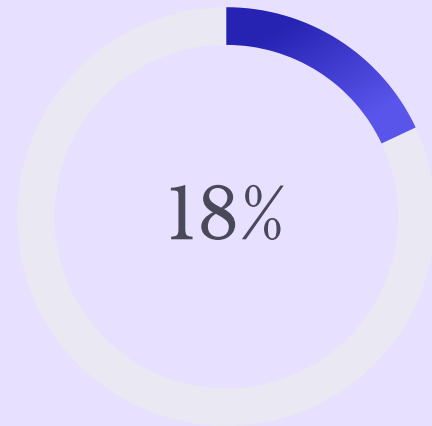
A major retailer was struggling with static recommendation systems that updated only once per day, resulting in:

- Outdated suggestions that didn't reflect browsing session
- Missed cross-selling opportunities during active sessions
- Low conversion rates compared to competitors

The Solution

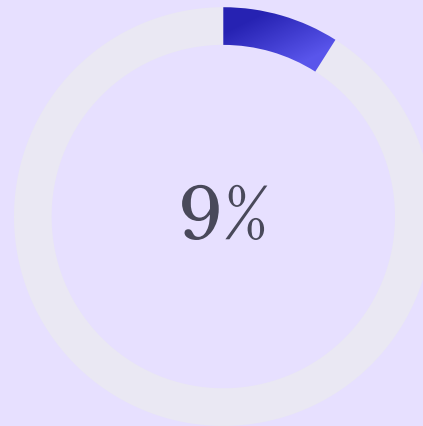
Implementation of real-time embedding-based personalization with:

- Kafka-powered event streaming
- Feast feature store for low-latency feature access
- Pinecone vector database for similarity search
- KServe for elastic model deployment



CTR Improvement

Increase in click-through rate on personalized recommendations



Revenue Per User

Growth in average revenue per user after implementation



Response Time

Average latency for generating personalized recommendations



Case Study: Bank Fraud Detection

⊗ The Problem

A global financial institution was processing fraud detection in batch, leading to:

- \$25M+ in annual fraud losses
- Poor customer experience from false positives
- Detection delays of hours after transactions

The Solution: Real-Time Fraud Scoring



70%

Fraud Reduction

Decrease in fraud losses after implementation

<40ms

Decision Time

Latency for complete transaction evaluation

Real-Time AI Infrastructure



Milliseconds matter in enterprise AI.

Modern real-time AI infrastructure combines streaming data processing, feature management, and optimized inference to deliver decisions in milliseconds—translating directly into competitive advantage and business value.

Summary: Real-Time AI Drives Enterprise Value

Business Impact

Real-time AI = **instant decisions = enterprise value**, with milliseconds directly translating to millions in revenue and cost savings

Key Domains

Advertising, fraud detection, and personalization represent the most mature and highest-impact applications of real-time AI technologies

Technical Foundation

Success requires robust integration of **streaming data, feature stores, and low-latency serving** infrastructure

Critical Trade-offs

Organizations must carefully balance latency requirements against model accuracy and infrastructure costs

Next Steps

Evaluate your current AI applications for real-time opportunities and prioritize use cases where millisecond improvements drive substantial business value.